



Guia – Email Server

Um servidor de e-mail completo no Linux requer pelo menos três componentes principais:

1. **Mail Transfer Agent (MTA):** Responsável por enviar e-mails de um servidor para outro (SMTP).
 - **Exemplos Comuns:** Postfix (o mais popular e recomendado), Exim, Sendmail.
2. **Mail Delivery Agent (MDA):** Recebe e armazena e-mails nos *mailboxes* dos usuários. Também lida com a recuperação via POP3/IMAP.
 - **Exemplos Comuns:** Dovecot (o mais popular), Cyrus, Courier.
3. **DNS (Domain Name System) Records:** Essenciais para que outros servidores de e-mail possam encontrar e validar seu servidor.
 - **Registros Chave:** MX (Mail Exchange) para direcionar o e-mail para seu servidor, A (endereço IP do servidor), e SPF/DKIM/DMARC para evitar que seus e-mails sejam marcados como *spam*.

1. Preparação e Instalação do Postfix (MTA)

Instalar o Postfix

1. Atualizar o Sistema:

```
sudo apt update  
sudo apt upgrade
```

2. Instalar Postfix e Mailutils:

```
sudo apt install postfix mailutils
```

- Durante a instalação, será solicitado o tipo de configuração. Escolha **"Internet Site"**.
- Quando solicitado o **"System mail name"** (Nome de e-mail do sistema), insira seu domínio principal (ex: seu-dominio.com).

3. Configurar o Postfix Principal:

- Edite o ficheiro de configuração principal do Postfix:

```
sudo nano /etc/postfix/main.cf
```



Defina ou verifique as seguintes configurações (substitua pelos seus valores):

- `myhostname = mail.seu-dominio.com` (Nome de host completo do seu servidor)
- `mydomain = seu-dominio.com` (domínio principal)
- `myorigin = $mydomain` (Domínio anexado a endereços de e-mail não qualificados)
- `mydestination = $myhostname, localhost.$mydomain, $mydomain` (Para que domínios este servidor aceita e-mail)
- `mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128` (Redes locais que têm permissão para retransmissão)
- `mailbox_size_limit = 0` (Permite caixas de correio de qualquer tamanho)
- `recipient_delimiter = +`
- `inet_protocols = all`
- `home_mailbox = Maildir/`

4. Reiniciar Postfix:

```
sudo systemctl restart postfix
```

2. Instalação e Configuração do Dovecot (MDA/IMAP/POP3)

O Dovecot permite que os utilizadores acessem às suas caixas de correio usando clientes de e-mail (Outlook, Thunderbird, etc) via POP3 ou IMAP.

1. Instalar Dovecot:

```
sudo apt install dovecot-imapd dovecot-pop3d
```

2. Configurar Dovecot para Maildir: Edite o ficheiro de configuração de caixas de correio:

```
sudo nano /etc/dovecot/conf.d/10-mail.conf
```

Altere a linha `mail_location` para usar o formato **Maildir**:

```
mail_location = maildir:~/Maildir
```

3. Reiniciar Dovecot:

```
sudo systemctl restart dovecot
```



4. Configuração de Utilizadores

Os utilizadores de e-mail são geralmente **utilizadores do sistema Linux**.

1. Criar um Novo Utilizador de E-mail:

```
sudo adduser nome_de_utilizador
```

- Isto cria um utilizador no sistema e um diretório *home* onde o Postfix e o Dovecot armazenarão os e-mails (especialmente se estiver usando Maildir).

2. Verificar a Criação do Maildir: Envie um e-mail de teste para o novo utilizador com o comando `mail` (requer `mailutils`):

```
echo "Teste de e-mail local." | mail -s "Teste Interno"  
nome_de_utilizador
```

O Dovecot criará automaticamente a estrutura Maildir (ex: `~/Maildir/cur`, `~/Maildir/new`, `~/Maildir/tmp`) na primeira entrega.

5. Utilize o cliente de email para se ligar ao servidor que acabou de criar/configurar

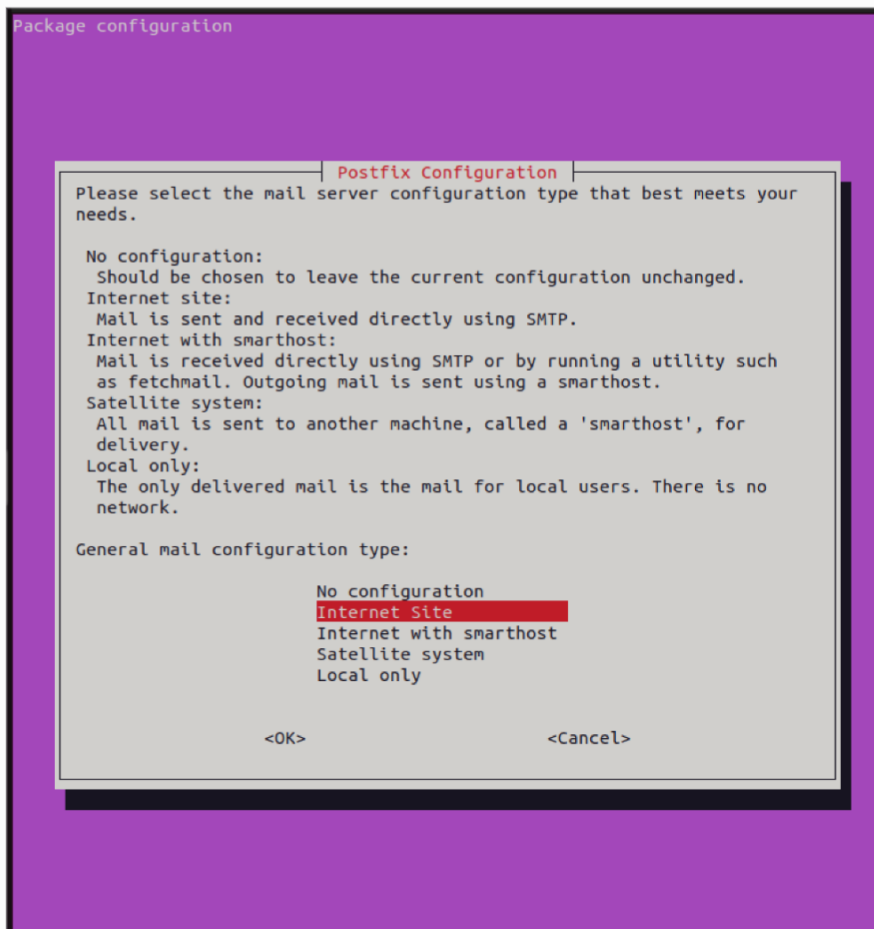
6. É possível que falte configurar o servidor DNS. Com base nos exercícios das aulas anteriores, configure o servidor DNS.

1. Preparação e Instalação do Postfix (MTA)

1.

```
ines@Ubuntu22:~$ sudo apt-get update
[sudo] password for ines:
Hit:4 http://security.ubuntu.com/ubuntu
Hit:1 https://pt.archive.ubuntu.com/ubuntu
Get:2 https://pt.archive.ubuntu.com/ubuntu
Get:3 https://pt.archive.ubuntu.com/ubuntu
Hit:5 https://ppa.launchpadcontent.net/
Get:6 https://pt.archive.ubuntu.com/ubuntu
[112 kB]
Get:7 https://pt.archive.ubuntu.com/ubuntu
tadata [212 B]
Get:8 https://pt.archive.ubuntu.com/ubuntu
data [359 kB]
Get:9 https://pt.archive.ubuntu.com/ubuntu
tadata [940 B]
Get:10 https://pt.archive.ubuntu.com/ubuntu
ata [7,144 B]
Get:11 https://pt.archive.ubuntu.com/ubuntu
Metadata [212 B]
Get:12 https://pt.archive.ubuntu.com/ubuntu
etadata [9,724 B]
Get:13 https://pt.archive.ubuntu.com/ubuntu
Metadata [212 B]
Fetched 745 kB in 1s (809 kB/s)
Reading package lists... Done
ines@Ubuntu22:~$ sudo apt-get upgrade
```

2.



Package configuration

Postfix Configuration

The 'mail name' is the domain name used to 'qualify' _ALL_ mail addresses without a domain name. This includes mail to and from <root>: please do not make your machine send out mail from root@example.org unless root@example.org has told you to.

This name will also be used by other programs. It should be the single, fully qualified domain name (FQDN).

Thus, if a mail address on the local host is foo@example.org, the correct value for this option would be example.org.

System mail name:

lr.estga.ua.pt

<OK>

<Cancel>

>> sudo apt purge postfix (comando para desinstalar e voltar a instalar o postfix caso precise)

```
GNU nano 6.2 /etc/bind/zones/lr.estga.ua.pt
$ORIGIN lr.estga.ua.pt.

$TTL      604800
@         IN      SOA      ns1r.lr.estga.ua.pt. email.do.gestor. (
                                1          ; Serial
                                7200       ; Refresh
                                120        ; Retry
                                2419200    ; Expire
                                604800)    ; Default TTL

@         IN      NS       ns1r.lr.estga.ua.pt.
@         IN      MX       10 mail.lr.estga.ua.pt.
ns1r      IN      A        192.168.1.1
Hostmachine IN      A        192.168.1.1
Debapp    IN      A        192.168.1.2
Win2008   IN      A        192.168.1.3
Elastix   IN      A        192.168.1.4
Zentyal   IN      A        192.168.1.5
pasg      IN      A        192.168.1.6
lmat      IN      A        192.168.1.254
mail      IN      A        192.168.1.4
www       IN      CNAME     mail.lr.estga.ua.pt
```

```

GNU nano 6.2 /etc/bind/zones/lr.estga.ua.pt *
$ORIGIN lr.estga.ua.pt.

$TTL      604800
@         IN      SOA      ns1r.l1r.estga.ua.pt. email.do.gestor. (
                                1          ; Serial
                                7200       ; Refresh
                                120        ; Retry
                                2419200    ; Expire
                                604800)    ; Default TTL

@         IN      NS       ns1r.l1r.estga.ua.pt.
@         IN      MX       10 mail.l1r.estga.ua.pt.
ns1r      IN      A        192.168.1.1
Hostmachine IN      A        192.168.1.1

Debapp    IN      A        192.168.1.2
Win2008   IN      A        192.168.1.3
Elastix   IN      A        192.168.1.7
Zentyal   IN      A        192.168.1.5
pasg      IN      A        192.168.1.6
lmat      IN      A        192.168.1.254

mail      IN      A        192.168.1.4
www       IN      CNAME    mail.l1r.estga.ua.pt

```

```

GNU nano 6.2 192.168.1.rev *
@         IN      SOA      ns1r.l1r.estga.ua.pt. admin.email.addr. (
                                2011101501 ; Serial
                                7200       ; Refresh
                                120        ; Retry
                                2419200    ; Expire
                                604800)    ; Default TTL

                                IN      NS       ns1r.l1r.estga.ua.pt.
1         IN      PTR      ns1r.l1r.estga.ua.pt.
1         IN      PTR      Hostmachine

                                IN      PTR      ns1r
1         IN      PTR      Debapp
2         IN      PTR      Win2008
3         IN      PTR      Elastix
7         IN      PTR      Zentyal
5         IN      PTR      pasg
6         IN      PTR

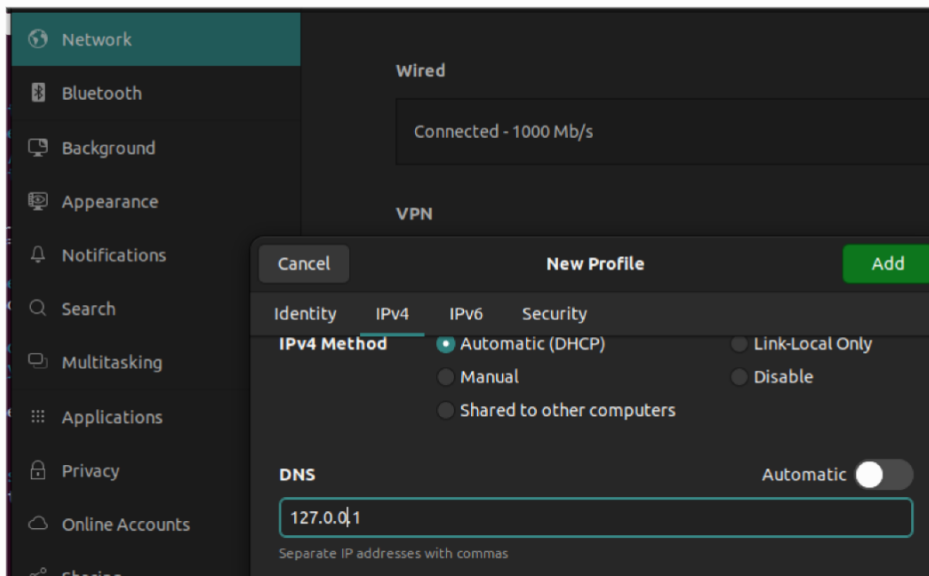
4         IN      PTR      mail.l1r.estga.ua.pt
254      IN      PTR      lmat

```

```

ines@Ubuntu22:~$ sudo systemctl restart named

```



3.

```
GNU nano 6.2 /etc/postfix/main.cf *
# appending .domain is the MUA's job.
append_dot_mydomain = no

# Uncomment the next line to generate "delayed mail" warnings
#delay_warning_time = 4h

readme_directory = no

# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 3.6 on
# fresh installs.
compatibility_level = 3.6

# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache

smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = Ubuntu22.04.myguest.virtualbox.org
myhostname = mail.lr.estga.ua.pt
mydomain = lr.estga.ua.pt
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
#myorigin = /etc/mailname
myorigin = $lr.estga.ua.pt
mydestination = $myhostname, lr.estga.ua.pt, Ubuntu22.04.myguest.virtualbox.org, localhost.04.myguest.virtualbox.org, localhost, localhost.$mydomain, $mydomain
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all
home_mailbox = Maildir/
```

4.

```
ines@Ubuntu22:~$ sudo systemctl restart postfix
```

2. Instalação e Configuração do Dovecot (MDA/IMAP/POP3)

1.

```
ines@Ubuntu22:~$ sudo apt install dovecot-imapd dovecot-pop3d
```

2.

```
ines@Ubuntu22:~$ sudo nano /etc/dovecot/conf.d/10-mail.conf
```

```
GNU nano 6.2 /etc/dovecot/conf.d/10-mail.conf *
##
## Mailbox locations and namespaces
##

# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
# mail_location = maildir:~/Maildir
# mail_location = mbox:~/mail:INBOX=/var/mail/%u
# mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
# <doc/wiki/MailLocation.txt>
#
#mail_location = mbox:~/mail:INBOX=/var/mail/%u
mail_location = maildir:~/Maildir

# If you need to set multiple mailbox locations or want to change default
# namespace settings, you can do it by defining namespace sections.
#
# You can have private, shared and public namespaces. Private namespaces
# are for user's personal mails. Shared namespaces are for accessing other
# users' mailboxes that have been shared. Public namespaces are for shared
# mailboxes that are managed by sysadmin. If you create any shared or public
# namespaces you'll typically want to enable ACL plugin also, otherwise all
```

3.

```
ines@Ubuntu22:~$ sudo systemctl restart dovecot
```


4. Configuração de Utilizadores

1. user inesns

password: 12345678

```
ines@Ubuntu22:~$ sudo adduser inesns
Adding user `inesns' ...
Adding new group `inesns' (1001) ...
Adding new user `inesns' (1001) with group `inesns' ...
Creating home directory `/home/inesns' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: password updated successfully
Changing the user information for inesns
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n]
```

2.

```
ines@Ubuntu22:~$ echo "Teste de email local." | mail -s "Teste Interno" inesns
ines@Ubuntu22:~$ cd ~/Maildir
ines@Ubuntu22:~/Maildir$ ll
total 20
drwx-----  5 ines ines 4096 Nov 28 12:45 ./
drwxr-x--- 20 ines ines 4096 Nov 28 12:45 ../
drwx-----  2 ines ines 4096 Nov 28 12:45 cur/
drwx-----  2 ines ines 4096 Nov 28 12:45 new/
drwx-----  2 ines ines 4096 Nov 28 12:45 tmp/
```

```
root@Ubuntu22:/home/novouser/Maildir/new# ll
total 12
drwx-----  2 novouser novouser 4096 Dec  5 12:41 ./
drwx-----  5 novouser novouser 4096 Dec  5 12:41 ../
-rw-----  1 novouser novouser  497 Dec  5 12:41 1764938478.V803I4000cM56143.Ubuntu22
root@Ubuntu22:/home/novouser/Maildir/new# cat 1764938478.V803I4000cM56143.Ubuntu22
Return-Path: <ines@Ubuntu22.04.myguest.virtualbox.org>
X-Original-To: novouser
Delivered-To: novouser@lr.estga.ua.pt
Received: by mail.lr.estga.ua.pt (Postfix, from userid 1000)
        id 095202A01FE; Fri,  5 Dec 2025 12:41:18 +0000 (GMT)
Subject: Teste Interno
To: novouser@lr.estga.ua.pt
User-Agent: mail (GNU Mailutils 3.14)
Date: Fri,  5 Dec 2025 12:41:18 +0000
Message-Id: <20251205124118.095202A01FE@mail.lr.estga.ua.pt>
From: ines <ines@Ubuntu22.04.myguest.virtualbox.org>

Teste de email local.
```

5. Utilize o cliente de email para se ligar ao servidor que acabou de criar/configurar:

```
root@Ubuntu22:/home/novouser# telnet localhost 25
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
220 mail.lr.estga.ua.pt ESMTP Postfix (Ubuntu)
```

```
ines@Ubuntu22:/home$ sudo ps aux | grep telnet
[sudo] password for ines:
root      9496  0.0  0.0  6280  3328 pts/1    S+   12:46   0:00 telnet localhost 25
ines      9519  0.0  0.0  9084  2560 pts/2    S+   12:48   0:00 grep --color=auto telnet
ines@Ubuntu22:/home$ sudo kill -9 9496
```

6. É possível que falte configurar o servidor DNS. Com base nos exercícios das aulas anteriores, configure o servidor DNS.

Já tinha configurado.